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# Cohort : July

# Date; 10/11/2026

**SOC Phishing Investigation Report PART A**

Four Email Investigation Summaries and Consolidated Indicator of Compromise (IOC) Register

Investigation methodology: Detect → Investigate → Enrich → Respond → Verify

Investigation objective: To analyse four suspicious emails, identify indicators of compromise (IOCs), establish the suspected objectives of the threat actors, assess the credibility of the evidence and recommend appropriate security actions.

**1. Individual email investigation summaries**

**Email 1: Microsoft 365 Account Verification – Brand Impersonation**

Malicious – High confidence

Suspected objective

The suspected objective is credential harvesting through Microsoft brand impersonation. The attacker attempts to deceive the recipient into visiting a fraudulent Microsoft 365 verification page and submitting their account credentials. The account suspension and verification theme is intended to create urgency and encourage immediate action.

Evidence identified

- The sender's Return-Path is noreply@mailer-srv-dispatch.xyz, while the Reply-To address uses ms-support@account-ms365-verify.top.
- The email uses PHPMailer 5.2.27, which is inconsistent with the claimed Microsoft identity.
- The originating IP address, 185.220.101.47, was reported as 100% critical by AbuseIPDB and associated with a Tor exit node.
- SPF, DKIM and DMARC authentication all failed.
- The email headers contain suspicious routing information, including helo=localhost.localdomain.
- The Microsoft verification URL uses the unrelated domain login-microsoft365-verify.top.
- The verification and unsubscribe URLs were not flagged by VirusTotal, but this does not establish that they are safe.

Verdict

The email is classified as a malicious phishing attempt involving brand impersonation and suspected credential harvesting. The combination of failed authentication, suspicious sender infrastructure, domain inconsistencies and a high-risk originating IP provides strong evidence of malicious activity.

Confidence: High (95%).

Limitations

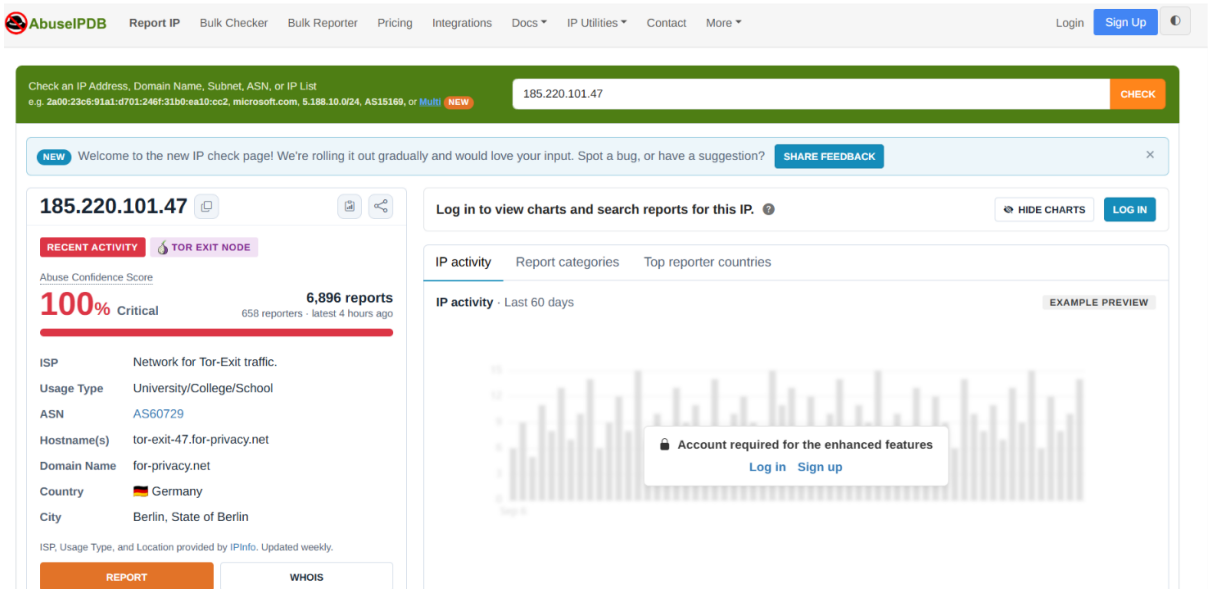
- The actual landing-page content and credential-submission functionality were not independently examined.
- The verification URL was not flagged by VirusTotal at the time of analysis.
- IP reputation data does not independently establish the identity of the attacker.
- The internal 127.0.0.1 address is a loopback address and should not be treated as an external attacker IP.

Recommended actions

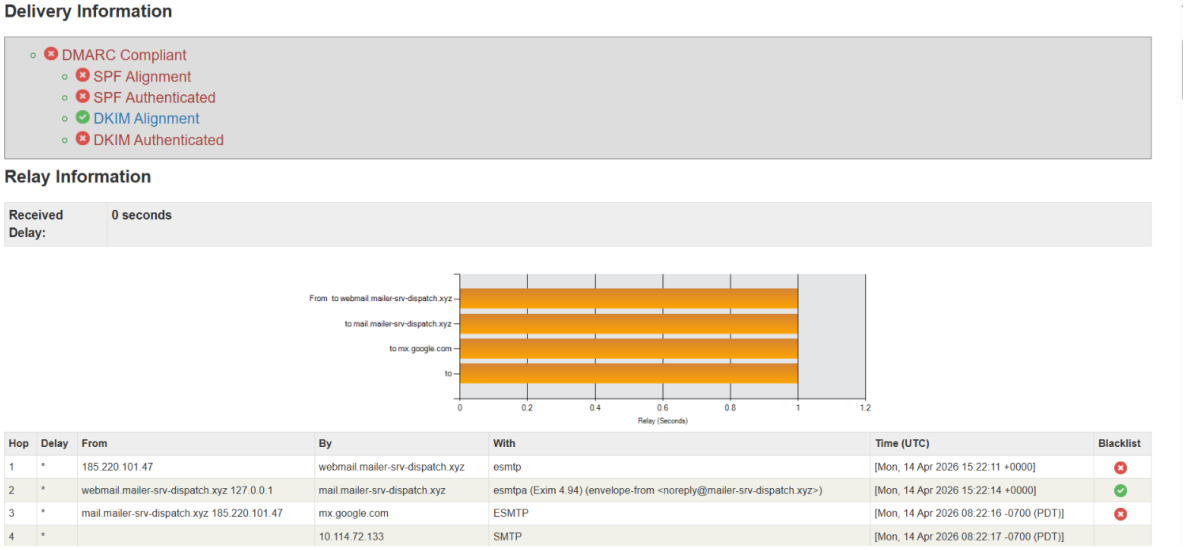
1. Quarantine the email and block the identified malicious domains and URLs.
2. Block or monitor the originating IP address 185.220.101.47.
3. Search email logs for other messages originating from the same infrastructure.
4. If credentials were submitted, immediately reset the affected account password and revoke active sessions.
5. Enforce MFA and investigate suspicious sign-in activity.
6. Add the phishing indicators to the organisation's SIEM threat-intelligence database.

   The PDF link decoded is shown below

MICROSOFT ACCOUNT VERIFICATION REQUIQED




Originating IP Address for Email1



**Delivery Information**

- DMARC Compliant
  - SPF Alignment
  - SPF Authenticated
  - DKIM Alignment
  - DKIM Authenticated

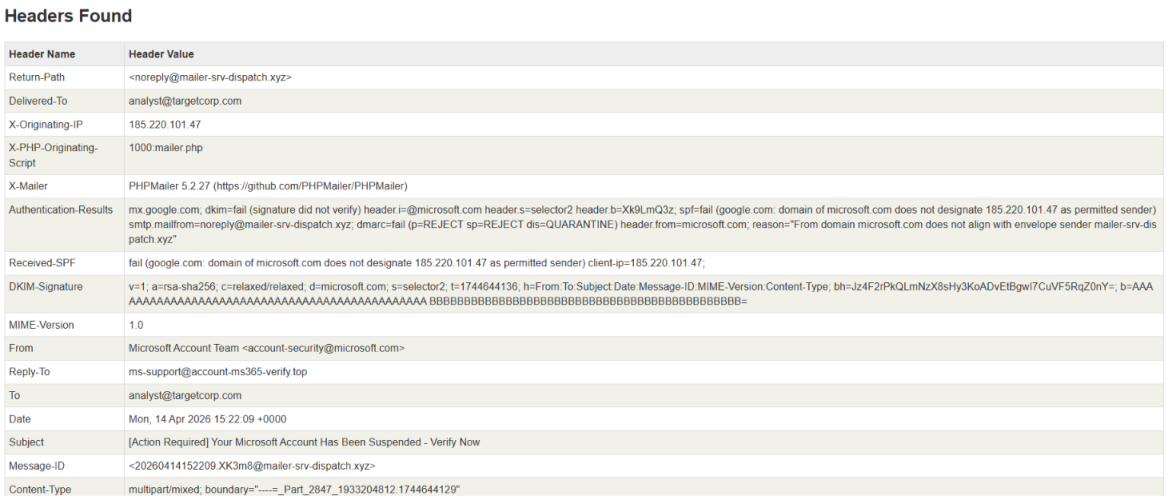
**Relay Information**

| | |
|---|---|
| Received Delay: | 0 seconds |



| Hop | Delay | From | By | With | Time (UTC) | Blacklist |
|---|---|---|---|---|---|---|
| 1 | * | 185.220.101.47 | webmail.mailer-srv-dispatch.xyz | esmtp | [Mon, 14 Apr 2026 15:22:11 +0000] | |
| 2 | * | webmail.mailer-srv-dispatch.xyz 127.0.0.1 | mail.mailer-srv-dispatch.xyz | esmtpa (Exim 4.94) (envelope-from <noreply@mailer-srv-dispatch.xyz>) | [Mon, 14 Apr 2026 15:22:14 +0000] | |
| 3 | * | mail.mailer-srv-dispatch.xyz 185.220.101.47 | mx.google.com | ESMTP | [Mon, 14 Apr 2026 08:22:16 -0700 (PDT)] | |
| 4 | * | | 10.114.72.133 | SMTP | [Mon, 14 Apr 2026 08:22:17 -0700 (PDT)] | |

Failed SPF DKIM AND DMARC Image



**Headers Found**

| Header Name | Header Value |
|---|---|
| Return-Path | <noreply@mailer-srv-dispatch.xyz> |
| Delivered-To | analyst@targetcorp.com |
| X-Originating-IP | 185.220.101.47 |
| X-PHP-Originating-Script | 1000:mailer.php |
| X-Mailer | PHPMailer 5.2.27 (https://github.com/PHPMailer/PHPMailer) |
| Authentication-Results | mx.google.com; dkim=fail (signature did not verify) header.i=@microsoft.com header.s=selector2 header.b=Xk9LmQ3z; spf=fail (google.com: domain of microsoft.com does not designate 185.220.101.47 as permitted sender) smtp.mailfrom=noreply@mailer-srv-dispatch.xyz; dmarc=fail (p=REJECT sp=REJECT dis=QUARANTINE) header.from=microsoft.com; reason="From domain microsoft.com does not align with envelope sender mailer-srv-dispatch.xyz" |
| Received-SPF | fail (google.com: domain of microsoft.com does not designate 185.220.101.47 as permitted sender) client-ip=185.220.101.47; |
| DKIM-Signature | v=1; a=rsa-sha256; c=relaxed/relaxed; d=microsoft.com; s=selector2; t=1744644136; h=From:To:Subject:Date:Message-ID:MIME-Version:Content-Type; bh=Jz4F2rPkQLmNzX8sHy3KoADvEtBgwI7CuVF5RqZ0nY=; b=AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA BBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBBB= |
| MIME-Version | 1.0 |
| From | Microsoft Account Team <account-security@microsoft.com> |
| Reply-To | ms-support@account-ms365-verify.top |
| To | analyst@targetcorp.com |
| Date | Mon, 14 Apr 2026 15:22:09 +0000 |
| Subject | [Action Required] Your Microsoft Account Has Been Suspended - Verify Now |
| Message-ID | <20260414152209.XK3m8@mailer-srv-dispatch.xyz> |
| Content-Type | multipart/mixed; boundary="----=_Part_2847_1933204812.1744644129" |

**Email 2: Delta Supplies – Urgent Vendor Invoice and Bank Details Change**

Malicious – High confidence

Suspected objective

The suspected objective is Business Email Compromise (BEC) or invoice-redirection fraud. The attacker attempts to manipulate the accounts payable team into transferring funds to fraudulent banking details by pretending to represent Delta Supplies Corp.

Evidence identified

- The subject line contains urgent payment language: INVOICE #DS-2026-04418 - Payment Due: April 22, 2026 – URGENT.
- The greeting, "Dear Accounts Payable Team", is generic and does not identify the recipient.
- The sender requests payment using recently changed banking details without evidence of independent verification.
- The company name is Delta Supplies Corp, while the beneficiary account name is Delta Supplies Corp LLC.
- The supplied banking information identifies First Eastern Bank in Riga, Latvia.
- The Return-Path is billing@fast-invoice-send.ru, while the Reply-To is payments@fast-invoice-send.ru.

- The message uses multiple sending infrastructure domains and IP addresses:
    - 194.165.16.101
    - 194.165.16.88
    - 10.0.0.14
- SPF, DKIM and DMARC authentication failed.
- The sending IP 194.165.16.101 was reported by 29 AbuseIPDB users for port scanning and brute-force activity.
- The second external IP, 194.165.16.88, was reported by eight users for suspicious activities.

Verdict

The email is classified as a suspected malicious BEC and payment-redirection attempt. The urgency, unexplained banking changes, beneficiary-name discrepancy, sender-domain mismatch and failed email authentication indicate a significant risk of financial fraud.

Confidence: High (95%).

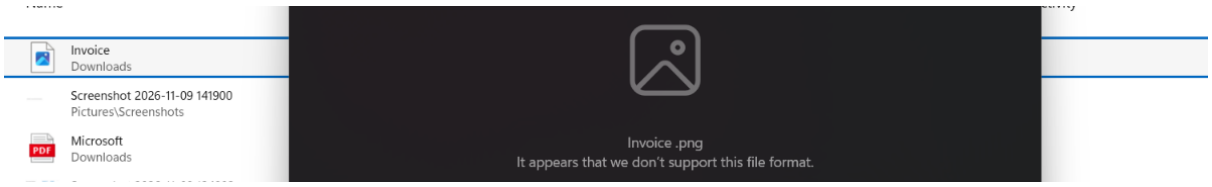
Limitations

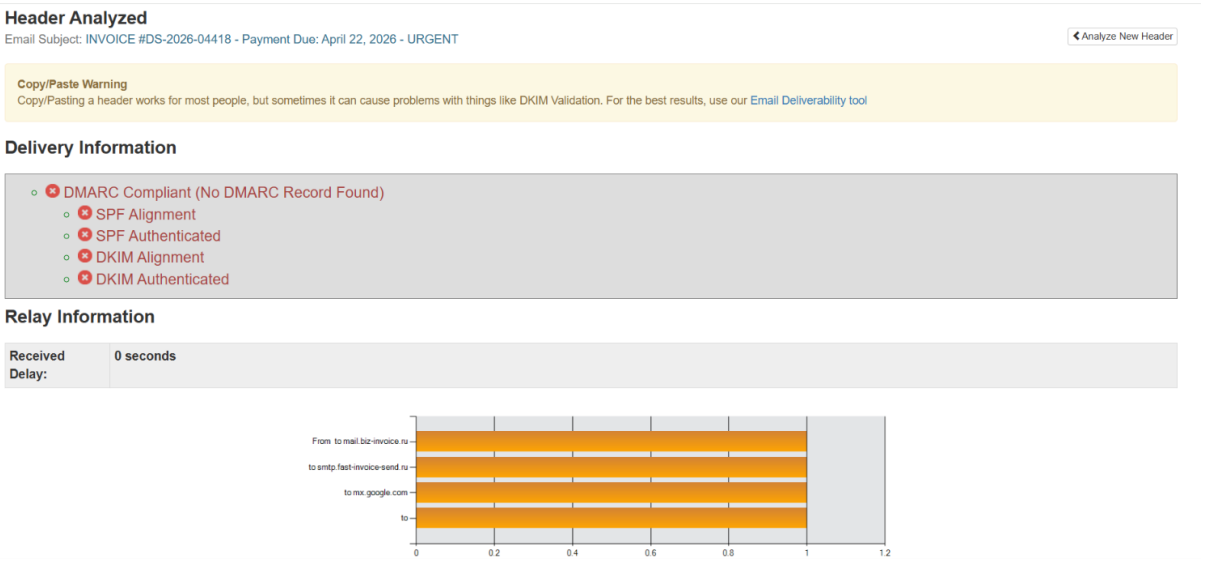
- The supplied bank account details have not been independently verified against the legitimate vendor.
- No evidence confirms whether the beneficiary account is controlled by the attacker.
- The contact email address was not flagged by VirusTotal, but this does not prove legitimacy.
- The low-risk IP reputation ratings do not eliminate the possibility of malicious use.

Recommended actions

1. Immediately suspend payment against invoice DS-2026-04418.
2. Contact Delta Supplies using previously verified contact information, not the details supplied in the email.
3. Independently verify any bank account changes using an established secondary communication channel.
4. Quarantine the email and block the sender domains.
5. Monitor accounts payable activity for attempted fraudulent transfers.
6. If payment has already been made, contact the organisation's bank immediately to request a payment recall or fraud investigation.
7. Search the SIEM and email gateway logs for related messages and indicators.

    INVOICE IMAGE EXTRACTED






**Email 3: DHL Shipping – Fake Delivery Redirection and Payment Request**

Malicious – High confidence

Suspected objective

The suspected objective is financial fraud and personal information harvesting. The attacker impersonates DHL and uses a fake parcel redelivery notification to persuade the recipient to provide their delivery address and pay a small administration fee of $2.99. This could expose personal information and payment card details.

Evidence identified

- The email uses a fake delivery notification and threatens that the parcel will be returned after five business days.
- The recipient is instructed to pay a $2.99 redelivery administration fee.
- The sender uses the domain dhl-delivery-info.xyz, which does not establish a legitimate DHL affiliation.
- The email contains multiple sending infrastructure addresses:
    - 176.111.174.55
    - 176.111.174.33
    - 172.16.88.9
- SPF, DKIM and DMARC authentication failed.
- The originating IP 176.111.174.55 was reported by four users for brute-force and port-scanning activity.
- The IP 176.111.174.33 was also associated with historical brute-force and port-scanning reports.
- VirusTotal flagged the following URL as suspicious:

https://dhl-delivery-info.xyz/reschedule

- The support page was also identified as a phishing link:

https://dhl-delivery-info.xyz/support

Verdict

The email is classified as a malicious DHL impersonation and payment-card phishing attempt. The suspicious domain, failed authentication, artificial delivery urgency and VirusTotal detections provide strong evidence of malicious intent.

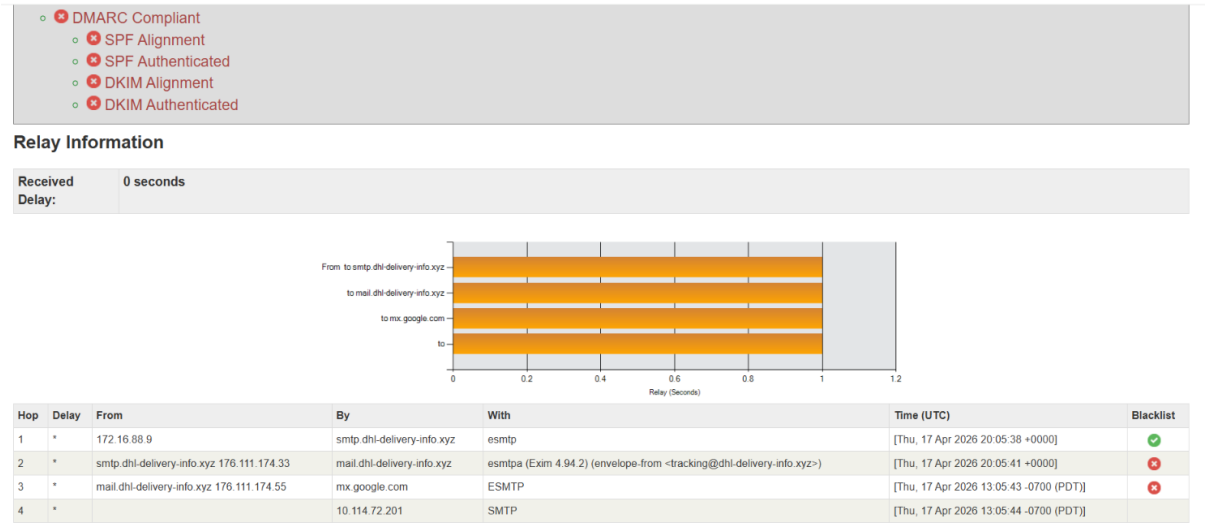
Confidence: High (98%).

Limitations

- The actual payment-processing functionality was not tested.
- No evidence confirms whether card information has already been collected.
- The internal address 172.16.88.9 cannot be attributed to an external attacker based on the supplied evidence.
- The IP reputation reports are historical and do not independently prove current malicious activity.

Recommended actions

1. Block the domain dhl-delivery-info.xyz and its associated URLs.
2. Quarantine the email and search for similar delivery-themed phishing messages.
3. Warn users not to submit personal information or payment details through the supplied links.
4. If card details were entered, contact the bank immediately to block or replace the card and review transactions.
5. If personal information was submitted, assess the potential for identity theft and targeted follow-up attacks.
6. Monitor the SIEM for connections to the identified domain and IP addresses.




**Email 4: Bank of America – Urgent Password Reset Alert**

Malicious – Very high confidence

Suspected objective

The suspected objective is credential theft and unauthorised access to the recipient's online banking account. The attacker impersonates Bank of America and pressures the recipient to complete an immediate password reset through a fraudulent website.

Evidence identified

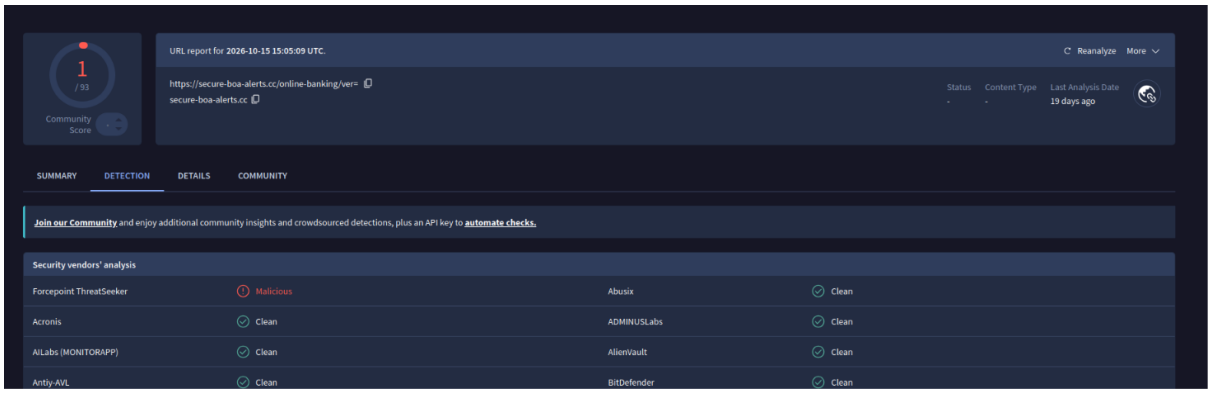
- The email uses the subject and message content to create urgency through an immediate password reset requirement.
- The Return-Path is noreply@secure-boa-alerts.cc, while the claimed identity is Bank of America.
- The sending domain secure-boa-alerts.cc is not the legitimate bank domain.
- The email contains multiple sending infrastructure addresses:

- 91.108.56.122
- 91.108.56.99
- 10.20.30.45

- SPF and DMARC authentication failed.
- DKIM failed because the reported RSA key size was only 512 bits, which is insufficient.
- The originating IP 91.108.56.122 was reported for port scanning and brute-force activity.
- The URL below was flagged by VirusTotal:

https://secure-boa-alerts.cc/online-banking/verify

- The domain and URL are inconsistent with the claimed Bank of America identity.




Flagged url




Verdict

The email is classified as a malicious banking phishing attempt with a suspected credential-harvesting objective. Failed email authentication, domain impersonation, a weak DKIM key and the VirusTotal detection make this a high-priority security threat.

Confidence: Very high (99%).

Limitations

- The actual phishing page and its credential-capture mechanism were not independently examined.

- The available evidence does not confirm whether the recipient interacted with the link.
- The IP 91.108.56.99 had no reported malicious activity in the supplied enrichment results; this does not establish that it is safe.
- The internal address 10.20.30.45 requires further investigation before attribution.

Recommended actions

1. Immediately block secure-boa-alerts.cc and the identified password-reset URL.
2. Quarantine the email and search for other messages using similar banking impersonation techniques.
3. If credentials were submitted, reset the password through the legitimate banking website and contact the bank.
4. Review account activity and investigate suspicious authentication attempts.
5. Enable MFA where available and revoke suspicious sessions.
6. Monitor the identified IP addresses and domain through SIEM detection rules.
7. Escalate the incident to the SOC senior analyst or incident response team.

**2. Consolidated IOC Register**

The following register combines the indicators extracted from the four investigated emails. It includes the observed reputation findings and their relevance to the investigation.

| Email | Indicator | Type | Enrichment findings | Assessment |
|---|---|---|---|---|
| 1 – Microsoft | 185.220.101.47 | IPv4 | AbuseIPDB: 100% critical; Tor exit node | High risk |
| 1 – Microsoft | 127.0.0.1 | IPv4 | Loopback address; reported abuse history supplied | Context required |
| 1 – Microsoft | mailer-srv-dispatch.xyz | Domain | Suspicious sending domain | Malicious context |
| 1 – Microsoft | account-ms365-verify.top | Domain | Microsoft impersonation domain | Malicious |
| 1 – Microsoft | login-microsoft365-verify.top | Domain | Used for fraudulent account verification | Suspicious |
| 1 – Microsoft | /signin/recover | URL | Not flagged by VirusTotal | Unconfirmed |
| 1 – Microsoft | /unsubscribe | URL | Not flagged by VirusTotal | Unconfirmed |
| 2 – Invoice | 194.165.16.101 | IPv4 | 29 reports; port scanning and brute force | Low reputation risk |
| 2 – Invoice | 194.165.16.88 | IPv4 | 8 reports; scanning, hacking and web attacks | Low reputation risk |
| 2 – Invoice | 10.0.0.14 | IPv4 | Private address; 3 brute-force reports | Context required |
| 2 – Invoice | fast-invoice-send.ru | Domain | Sender infrastructure; authentication failures | Suspicious |
| 2 – Invoice | biz-invoice.ru | Domain | Intermediate sending infrastructure | Suspicious |
| 2 – Invoice | billing@fast-invoice-send.ru | Email | Return-Path sender | Suspicious |
| 2 – Invoice | payments@fast-invoice-send.ru | Email | Reply-To address | Suspicious |
| 3 – DHL | 176.111.174.55 | IPv4 | 4 reports; brute force and port scanning | Low reputation risk |
| 3 – DHL | 176.111.174.33 | IPv4 | Historical brute-force and port-scanning reports | Low reputation risk |

| | | | | |
|---|---|---|---|---|
| 3 – DHL | 172.16.88.9 | IPv4 | Private address; no reports | Unconfirmed |
| 3 – DHL | dhl-delivery-info.xyz | Domain | DHL impersonation domain | Malicious context |
| 3 – DHL | /reschedule | URL | Flagged by VirusTotal | Malicious |
| 3 – DHL | /support | URL | Flagged as a phishing link | Malicious |
| 4 – Bank | 91.108.56.122 | IPv4 | 2 reports; port scanning and brute force | Low reputation risk |
| 4 – Bank | 91.108.56.99 | IPv4 | No reported abuse; requires further assessment | Unconfirmed |
| 4 – Bank | 10.20.30.45 | IPv4 | Private address; one report | Context required |
| 4 – Bank | secure-boa-alerts.cc | Domain | Bank of America impersonation domain | Malicious |
| 4 – Bank | /online-banking/verify | URL | Flagged by VirusTotal | Malicious |

Note: The table reflects the enrichment findings supplied in the capstone investigation. A low-risk or unreported indicator is not automatically benign. Private and loopback addresses should not be treated as publicly attributable attacker infrastructure.

**3. Consolidated Email Authentication Findings**

| Email | SPF | DKIM | DMARC | Overall assessment |
|---|---|---|---|---|
| Microsoft | Fail | Fail | Fail | Malicious |
| Delta Supplies invoice | Fail | Fail | Fail | Malicious |
| DHL delivery | Fail | Fail | Fail | Malicious |
| Bank of America | Fail | Fail | Fail | Malicious |

All four emails failed the reported SPF, DKIM and DMARC checks. These failures indicate that the messages could not establish the claimed sender identity through the supplied authentication results.

**4. Overall SOC Investigation Assessment**

Investigation outcome

**4**

Suspicious emails investigated

**4**

Emails classified as malicious

**12**

SPF/DKIM/DMARC failures across four emails

**4**

Distinct suspected attack objectives

The investigation identified four different phishing techniques targeting the organisation:

1. Microsoft impersonation: Credential harvesting through a fake Microsoft 365 verification page.
2. Invoice fraud: Business Email Compromise involving fraudulent vendor banking details and payment redirection.
3. DHL impersonation: Personal information and payment-card harvesting through a fake delivery redirection request.
4. Bank of America impersonation: Credential theft through an urgent password-reset notification.

**5. Final recommendations and incident response priorities**

| Priority | Recommended SOC action | Purpose |
|---|---|---|
| Critical | Quarantine all four emails | Prevent further user interaction |
| Critical | Block confirmed malicious domains and URLs | Prevent access to phishing infrastructure |
| Critical | Investigate affected user accounts | Identify possible credential compromise |
| Critical | Contact the finance team regarding the invoice | Prevent fraudulent financial transfers |
| High | Search SIEM and email gateway logs | Identify other affected users and related activity |
| High | Review authentication and network logs | Identify suspicious access and outbound connections |
| High | Reset compromised passwords and revoke sessions | Contain potential account compromise |
| Medium | Update email security and SIEM detection rules | Improve future phishing detection |
| Medium | Deliver targeted security awareness training | Reduce successful phishing attempts |

**6. Investigation limitations and verification requirements**

Although the investigation provides strong evidence of phishing, some additional verification is necessary before considering the incidents fully resolved.

- Preserve the original emails, including complete headers, as forensic evidence.
- Do not assume that different Received entries represent different malicious actors; they may represent successive mail-routing hops.
- Confirm the actual public source IPs using trusted header analysis.
- Investigate private IP addresses only in the context of the organisation's internal network.
- Recheck URLs and IP reputation against current threat-intelligence sources.
- Confirm whether any recipients clicked the links, submitted credentials, disclosed payment information or transferred funds.
- Verify that blocking and containment measures have been effective through subsequent SIEM and email gateway monitoring.

Final SOC conclusion

Based on the available evidence, all four emails should be treated as malicious and escalated for incident response. The investigation identified indicators of brand impersonation, credential harvesting, payment redirection and financial fraud.

The recommended response is to quarantine the messages, block confirmed malicious infrastructure, investigate potential user exposure and monitor the organisation's environment for related activity.

The SOC should only close each incident after confirming that containment has been successful, affected accounts and systems have been secured, and no further malicious activity has been detected.

# PART B:WAZUH DETECTION EXERCISES REPORT

*Wazuh – YARA Malware Detection, Suspicious Binary Detection and Shellshock*

Test date: 6–9 November 2026 | Environment: Kali Linux test host and Ubuntu endpoint

## 1. Executive summary

The supplied project evidence supports three investigation findings. First, Wazuh file-integrity monitoring recorded changes in the YARA malware test directory, including the Mirai-labelled file and a before/after SHA-256 difference. Second, Wazuh Rootcheck detected a trojaned /usr/bin/w binary on Ubuntu2. Third, Wazuh generated a Shellshock attack alert after an HTTP GET request from the Kali host to the Ubuntu web server included a Bash function payload followed by a command to read /etc/passwd.

The evidence confirms detection events. It does not, by itself, prove successful malware execution, data exfiltration, persistence, or complete remediation. The restoration-related screenshots show file deletion, integrity changes and a subsequent PAM login event, but do not explicitly demonstrate that /usr/bin/w was replaced with a verified clean package copy.

## 2. Environment and pre-test health check

The screenshots show a Wazuh-managed Ubuntu endpoint and a separate Kali Linux test host. The Kali terminal shows interface eth0 configured as 192.168.1.53. The Shellshock alert identifies the Ubuntu target as 192.168.1.52 (agent name Ubuntu2, agent ID 001). A separate Wazuh overview screenshot shows an active Ubuntu agent. The Ubuntu health-check screenshots include Ubuntu system information; the visible OS-release output identifies Ubuntu 24.04 LTS. The Kali OS screenshot identifies Kali GNU/Linux Rolling.

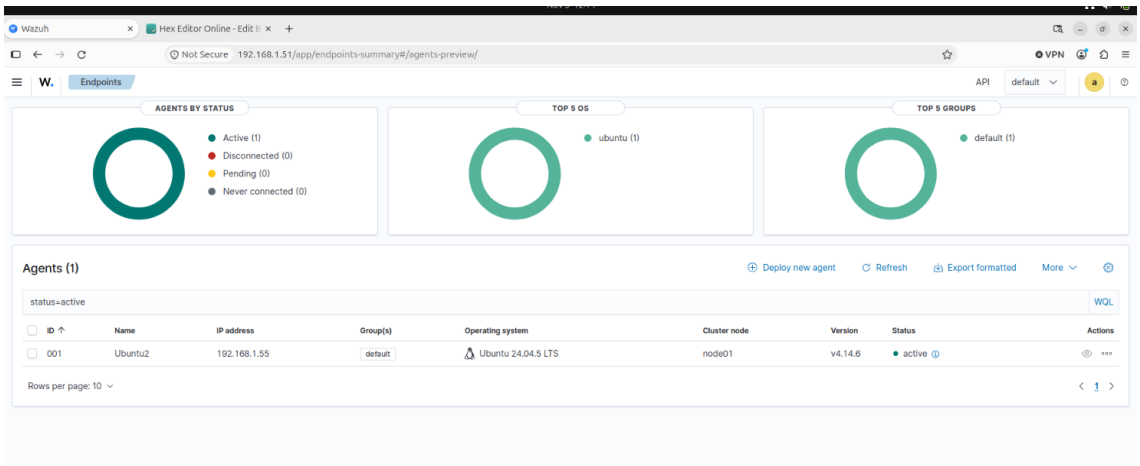



Figure 1. Wazuh dashboard showing the Ubuntu agent active.

```
ubuntu@ubuntu-VirtualBox:~/Desktop$ cat /etc/os-release
PRETTY_NAME="Ubuntu 24.04.5 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.5 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
ubuntu@ubuntu-VirtualBox:~/Desktop$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:ce:96:ec brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.55/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
       valid_lft 6456sec preferred_lft 6456sec
    inet6 fe80::a00:27ff:fece:96ec/64 scope link
       valid_lft forever preferred_lft forever
ubuntu@ubuntu-VirtualBox:~/Desktop$ host name
Host name.home.arpa not found: 2(SERVFAIL)
ubuntu@ubuntu-VirtualBox:~/Desktop$ hostname
ubuntu-VirtualBox
ubuntu@ubuntu-VirtualBox:~/Desktop$
```

Figure 2. Ubuntu health-check / operating-system information captured before testing.

```
ubuntu@ubuntu-VirtualBox: ~/Desktop
ubuntu@ubuntu-VirtualBox:~/Desktop$ lscpu
Architecture:                         x86_64
  CPU op-mode(s):                     32-bit, 64-bit
  Address sizes:                      48 bits physical, 48 bits virtual
  Byte Order:                         Little Endian
CPU(s):                               1
  On-line CPU(s) list:                0
Vendor ID:                            AuthenticAMD
  Model name:                         AMD Ryzen 7 5700U with Radeon Graphics
    CPU family:                       23
    Model:                            104
    Thread(s) per core:               1
    Core(s) per socket:               1
    Socket(s):                        1
    Stepping:                         1
    BogoMIPS:                         3593.14
    Flags:                            fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clflush mmx fxsr sse sse2 ht syscall nx mmxext fxsr_opt rdtscp lm constant_tsc rep_
                                      good nopl xtopology nonstop_tsc cpuid extd_apicid tsc_known_freq pni pclmulqdq ssse3 fma cx16 sse4_1 sse4_2 movbe popcnt aes xsave avx f16c rdrand hypervis
                                      or lahf_lm cr8_legacy abm sse4a misalignsse 3dnowprefetch ssbd vmmcall fsgsbase bmi1 avx2 bmi2 rdseed adx clflushopt sha_ni arat
Virtualization features:
  Hypervisor vendor:                  KVM
  Virtualization type:                full
NUMA:
  NUMA node(s):                       1
  NUMA node0 CPU(s):                  0
Vulnerabilities:
  Gather data sampling:               Not affected
  Ghostwrite:                         Not affected
  Indirect target selection:          Not affected
  Itlb multihit:                      Not affected
  L1tf:                               Not affected
  Mds:                                Not affected
  Meltdown:                           Not affected
  Mmio stale data:                    Not affected
  Old microcode:                      Not affected
  Reg file data sampling:             Not affected
  Retbleed:                           Mitigation; untrained return thunk; SMT disabled
  Spec rstack overflow:               Mitigation; SMT disabled
```

Figure 3. Ubuntu hardware and system information from the pre-test check.

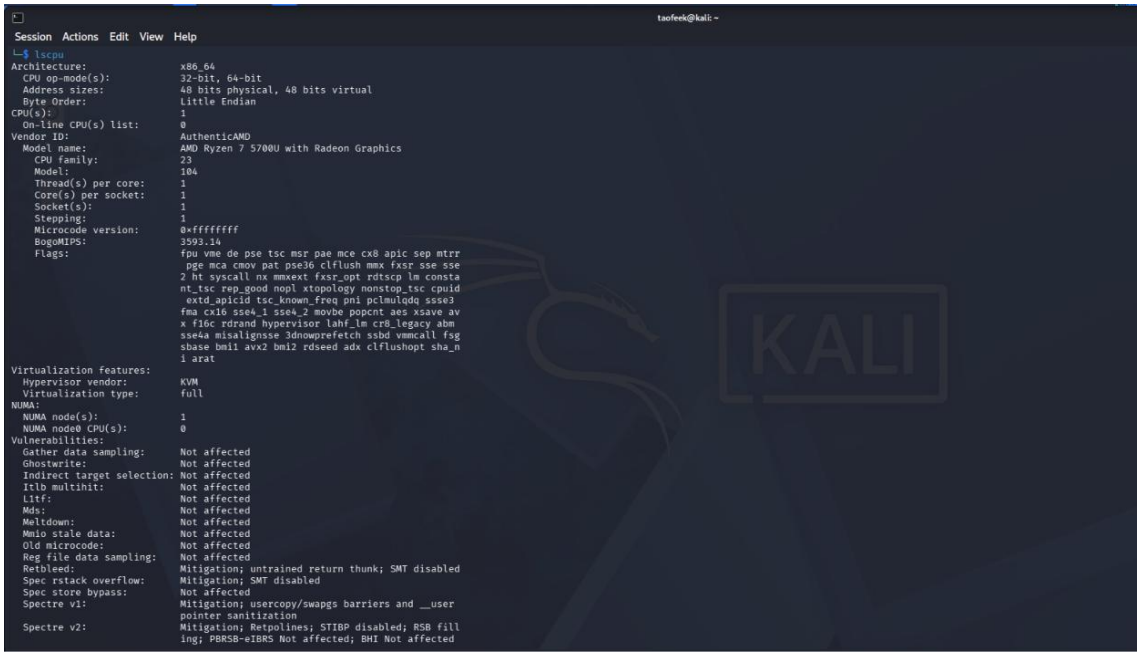



Figure 4. Additional Linux endpoint information captured during baseline checks.

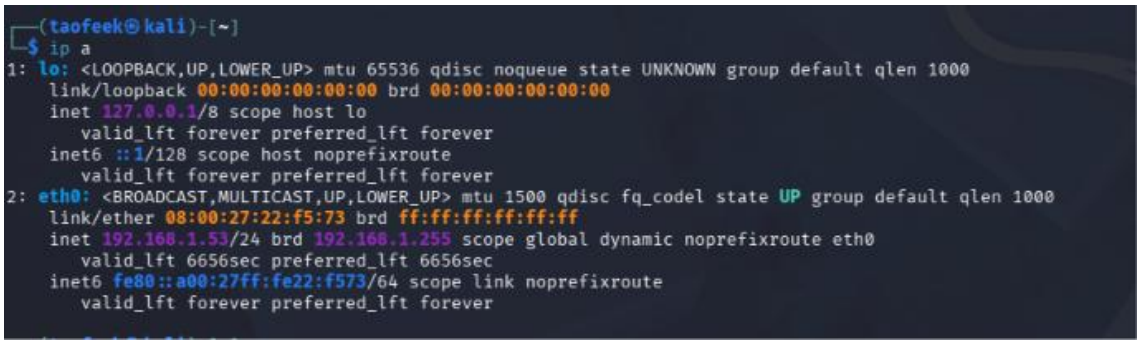



Figure 5. Kali test host network configuration; eth0 address 192.168.1.53.

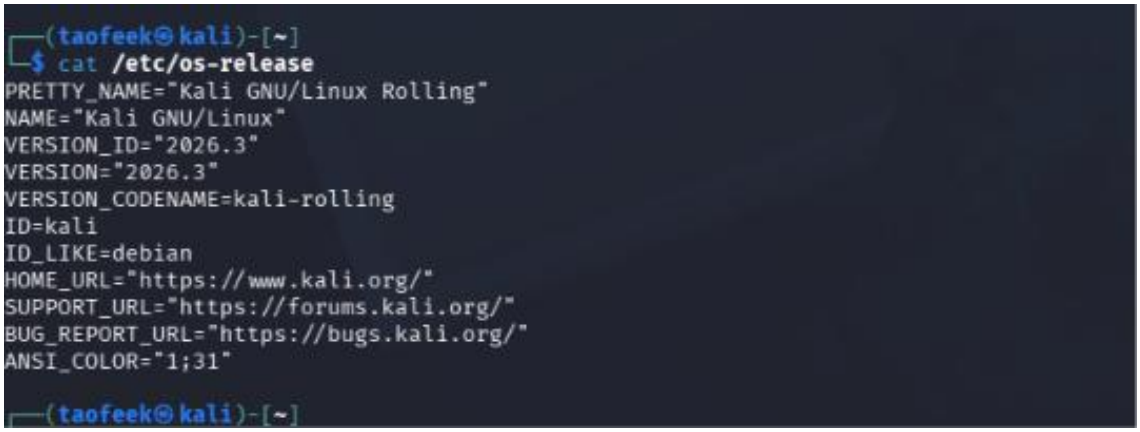



Figure 6. Kali operating-system release: Kali GNU/Linux Rolling.

# 3. Investigation 1 – YARA malware detection / file-integrity monitoring

## Objective and detection

The objective was to test Wazuh integration with YARA and identify changes to files in the malware test directory. The supplied alert records a modification under /tmp/yara/malware/ and identifies the sample path as /tmp/yara/malware/mirai.

## Supporting evidence

| Field | Recorded evidence |
|---|---|
| **Endpoint / agent** | Ubuntu2 (Wazuh agent shown in the screenshots) |
| **File path** | /tmp/yara/malware/mirai |
| **SHA-256 before** | f69560fda5f15a0a8693208171720cc93643e8de1001f227d7fcc893465df1e1 |
| **SHA-256 after** | 2aa2dbae345938386d93ed02d4f3aa3ad1b130c04b04e4e5c5c0d0e4ffe3181 |
| **Rule description** | File modified in /tmp/yara/malware/ directory |
| **Timestamp** | 6 November 2026, 20:44:49.823 |
| **Wazuh evidence** | File-integrity monitoring screenshots show added/modified file events in /tmp/yara/malware/ |

The different before/after SHA-256 values confirm that the file content changed. The Wazuh File Integrity Monitoring view also shows activity in the same test directory, including entries named webshell, vpn_filter and xbash, with file-added and file-modified events around 20:44:54–20:44:55 on 6 November 2026. These are supporting directory activity, but the screenshot does not show a YARA rule name or the exact matched rule output for the Mirai file.

## Suspected objective

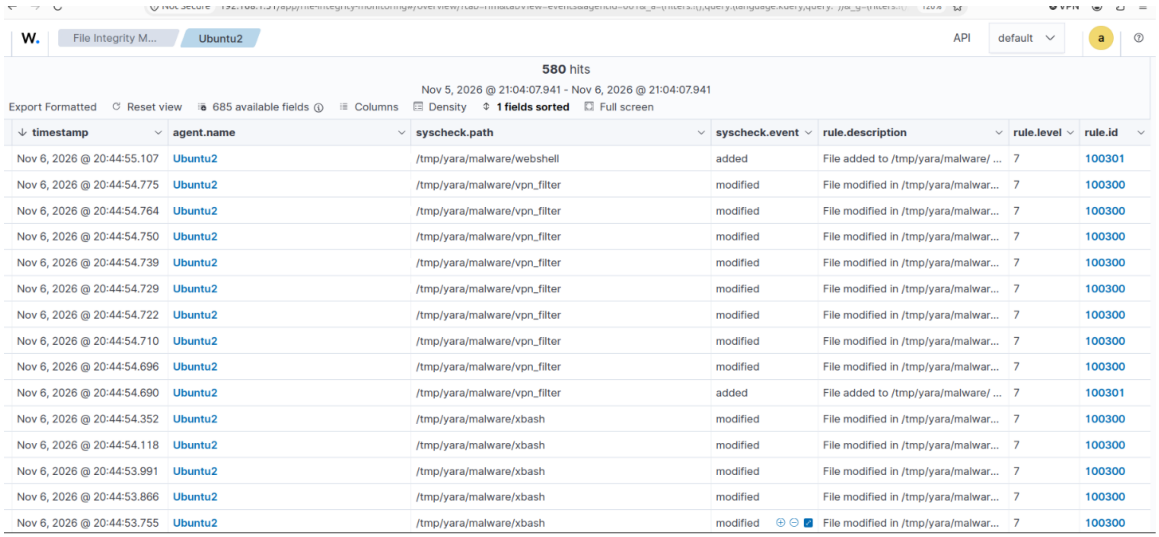
The activity is consistent with a controlled malware-detection exercise in which a Mirai-labelled sample or test artefact was introduced or modified to validate file-integrity/YARA monitoring. The filename alone does not prove that the sample is genuine Mirai malware or that it executed.

## Verdict, confidence and limitations

Verdict: File modification confirmed; malicious classification and execution are not independently proven by the supplied event alone. Confidence: High for the fact of modification because the before/after hashes differ and Wazuh recorded the event; moderate/undetermined for malware attribution. Limitation: the exact YARA match output, rule identifier, process-execution evidence and post-test scan result were not supplied.

## Recommended action

Preserve the sample and its hashes; quarantine it in the lab; retain the YARA rule and scan output; inspect the file in an isolated analysis environment; review process/audit logs for execution; and perform a fresh YARA scan after cleanup. Record the final hash and scan result before closing the test.




| timestamp | agent.name | syscheck.path | syscheck.event | rule.description | rule.level | rule.id |
|---|---|---|---|---|---|---|
| Nov 6, 2026 @ 20:44:55.107 | Ubuntu2 | /tmp/yara/malware/webshell | added | File added to /tmp/yara/malware/ ... | 7 | 100301 |
| Nov 6, 2026 @ 20:44:54.775 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.764 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.750 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.739 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.729 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.722 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.710 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.696 | Ubuntu2 | /tmp/yara/malware/vpn_filter | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.690 | Ubuntu2 | /tmp/yara/malware/vpn_filter | added | File added to /tmp/yara/malware/ ... | 7 | 100301 |
| Nov 6, 2026 @ 20:44:54.352 | Ubuntu2 | /tmp/yara/malware/xbash | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:54.118 | Ubuntu2 | /tmp/yara/malware/xbash | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:53.991 | Ubuntu2 | /tmp/yara/malware/xbash | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:53.866 | Ubuntu2 | /tmp/yara/malware/xbash | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |
| Nov 6, 2026 @ 20:44:53.755 | Ubuntu2 | /tmp/yara/malware/xbash | modified | File modified in /tmp/yara/malwar... | 7 | 100300 |

Figure 7. Wazuh FIM events for files in /tmp/yara/malware/ around the test time.

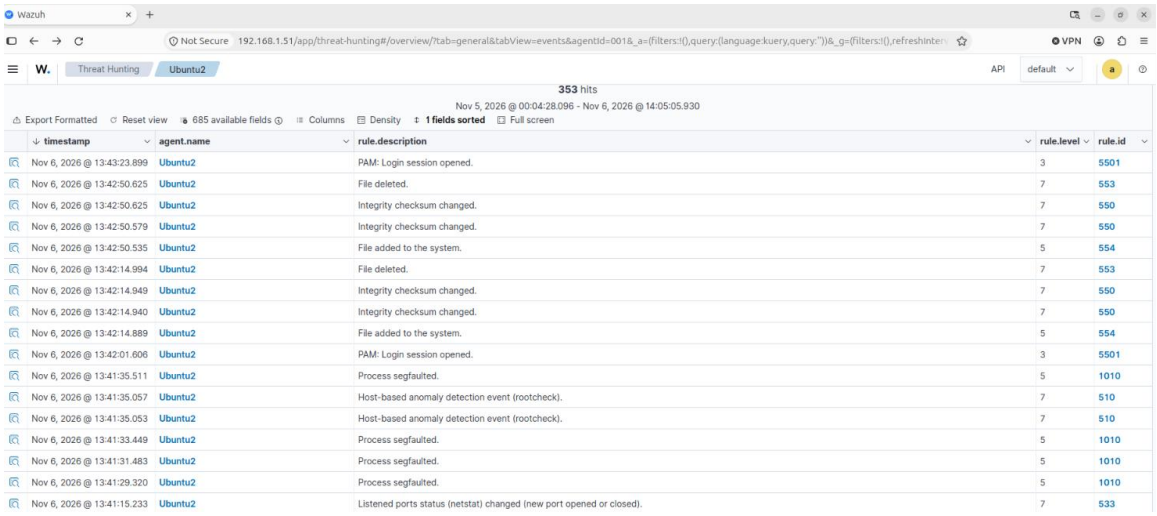



Figure 8. Wazuh event timeline showing file and process events on Ubuntu2.

# 4. Investigation 2 – Suspicious / trojaned system binary

## Objective and detection

The objective was to identify suspicious or modified Linux system binaries. Wazuh Rootcheck detected a trojaned version of /usr/bin/w on the Ubuntu2 endpoint. The event was recorded at 6 November 2026, 13:41:35.053. The generic signature was 'uname -a|proc\\.h|bash'.

## Supporting logs and alert evidence

| Field | Recorded evidence |
|---|---|
| **Agent** | Ubuntu2 |
| **Agent IP** | 192.168.1.55 |
| **Detected file** | /usr/bin/w |
| **Decoder / location** | rootcheck |
| **Rule description** | Host-based anomaly detection event (rootcheck) |
| **Rule ID / level** | 510 / 7 |
| **Timestamp** | 6 November 2026, 13:41:35.053 |
| **Signature** | uname -a\|proc\\.h\|bash |

The alert detail screenshot states: 'Trojaned version of file /usr/bin/w detected'. The accompanying project description says the trojan ran uname -a for system reconnaissance and a Bash script. The event timeline also shows process-segfault events at 13:41:31.483, 13:41:29.320 and 13:41:33.449, followed by the Rootcheck event at 13:41:35.053/13:41:35.057. PAM login-session events appear at 13:41:36.511 and 13:43:23.899. The logs establish temporal correlation, not by themselves a definitive causal chain.

## Suspected objective

The suspected objective was to disguise malicious commands within a legitimate system executable so that the binary could perform reconnaissance and launch Bash commands while appearing to be a normal utility.

## Verdict, confidence and limitations

Verdict: Wazuh detected a trojaned/suspicious /usr/bin/w binary. Confidence: High that Rootcheck generated the detection, based on the explicit alert, agent details and signature. The supplied evidence does not prove the full scope of compromise, the source of the binary modification, or successful execution of every command. The generic signature is an indicator, not a complete forensic reconstruction.

## Recommended action

Isolate the endpoint if it is not already contained; preserve a forensic copy and hash of /usr/bin/w; verify its ownership, permissions and package integrity; restore it from a trusted Ubuntu package source if tampering is confirmed; inspect related binaries, process execution and authentication logs; then run Rootcheck and file-integrity checks again. Do not treat the endpoint as clean until the restored binary has been verified.

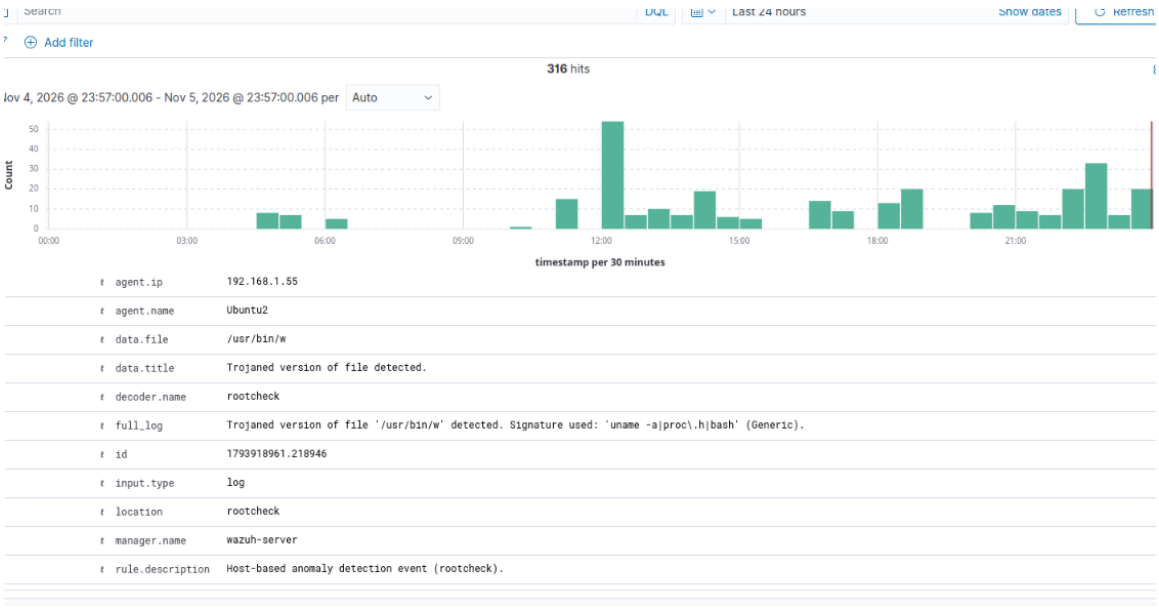



Figure 9. Wazuh Rootcheck alert details for the trojaned /usr/bin/w binary.




Figure 10. Event detail showing rule ID 510, level 7 and timestamp 13:41:35.053.

# 5. Investigation 3 – Shellshock attack-pattern detection

## Objective and detection

The objective was to validate Wazuh detection of a Shellshock-style payload sent to the Ubuntu web server. The Wazuh alert screenshot identifies agent Ubuntu2 (ID 001), target IP 192.168.1.52, source IP 192.168.1.53, HTTP GET to '/', and the alert description 'Shellshock attack detected'.

## Supporting logs

| Field | Recorded evidence |
|---|---|
| **Alert timestamp** | 9 November 2026, 13:32:24 (+0100) in the web access log |
| **Wazuh event timestamp** | 9 November 2026, 13:32:26.863 in the event list |
| **Target agent / IP** | Ubuntu2 / 192.168.1.52 |
| **Source IP** | 192.168.1.53 (Kali test host) |
| **HTTP request** | GET / |

| | |
|---|---|
| **HTTP status** | 200 |
| **Payload** | () { :; }; /bin/cat /etc/passwd |
| **Log source** | /var/log/apache2/access.log |
| **Decoder** | web-accesslog |
| **Rule description** | Shellshock attack detected |
| **Rule ID / level** | 31168 / 15 |

The web access log records the payload in the request and an HTTP 200 response. This proves that the request reached the web server and that the server returned status 200; it does not prove that the embedded command was executed successfully or that /etc/passwd contents were returned to the requester. The Wazuh event overview also shows the Shellshock alert at 13:32:26.863 and a later listened-port status change at 13:33:43.329.

## Suspected objective

The suspected objective was to exploit a vulnerable Bash invocation path through an HTTP request and execute an operating-system command. The command in the payload attempts to read /etc/passwd, which is consistent with basic host reconnaissance or proof-of-execution testing.

## Verdict, confidence and limitations

Verdict: Shellshock attack-pattern alert confirmed. Confidence: High that the test request matched the Wazuh Shellshock detection rule, supported by the source/target addresses, payload, log path and rule ID. Successful exploitation remains unconfirmed because the evidence does not include command output, process audit records or proof that file contents were disclosed.

## Recommended action

Patch or otherwise mitigate vulnerable Bash versions; review Apache and system logs for related requests; check process/audit telemetry for unexpected child processes; restrict access to exposed services; and repeat the test only in the authorised lab after confirming the detection and mitigation state.

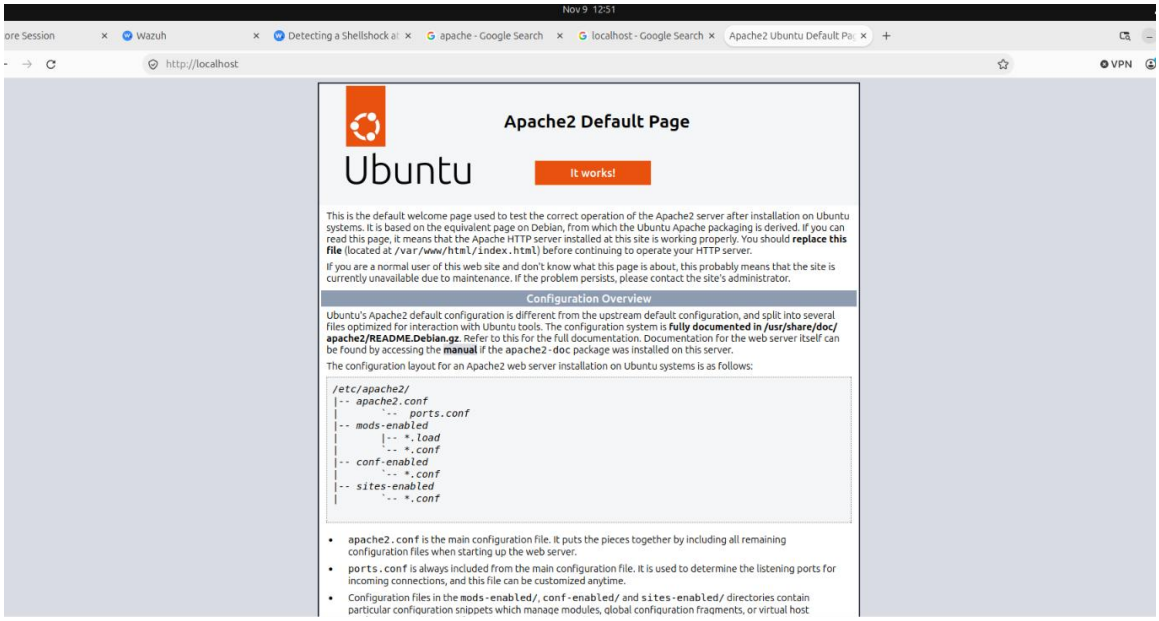



Figure 11. Web-server page shown during the Shellshock proof-of-concept test.

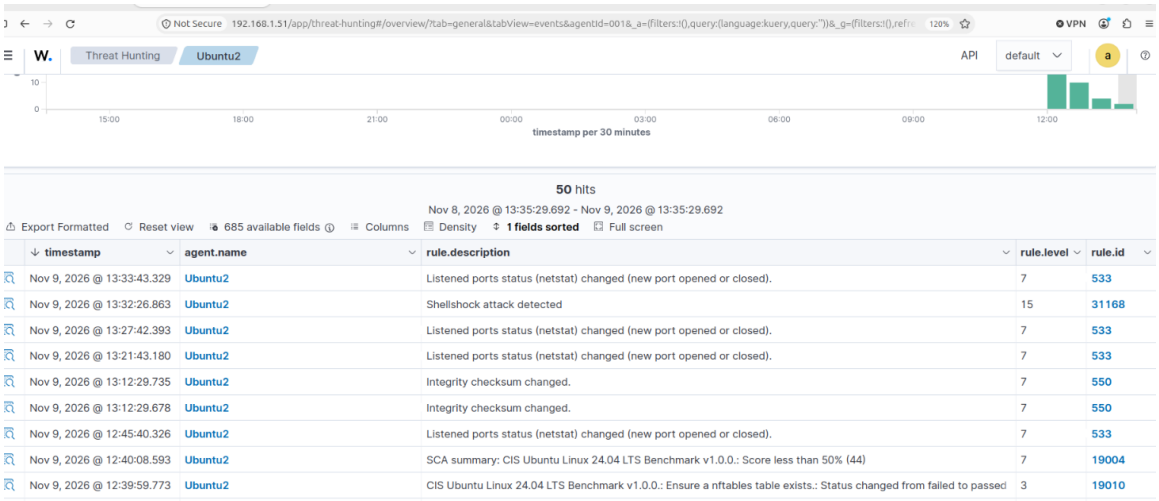



Figure 12. Wazuh threat-hunting event list showing the Shellshock alert and adjacent events.

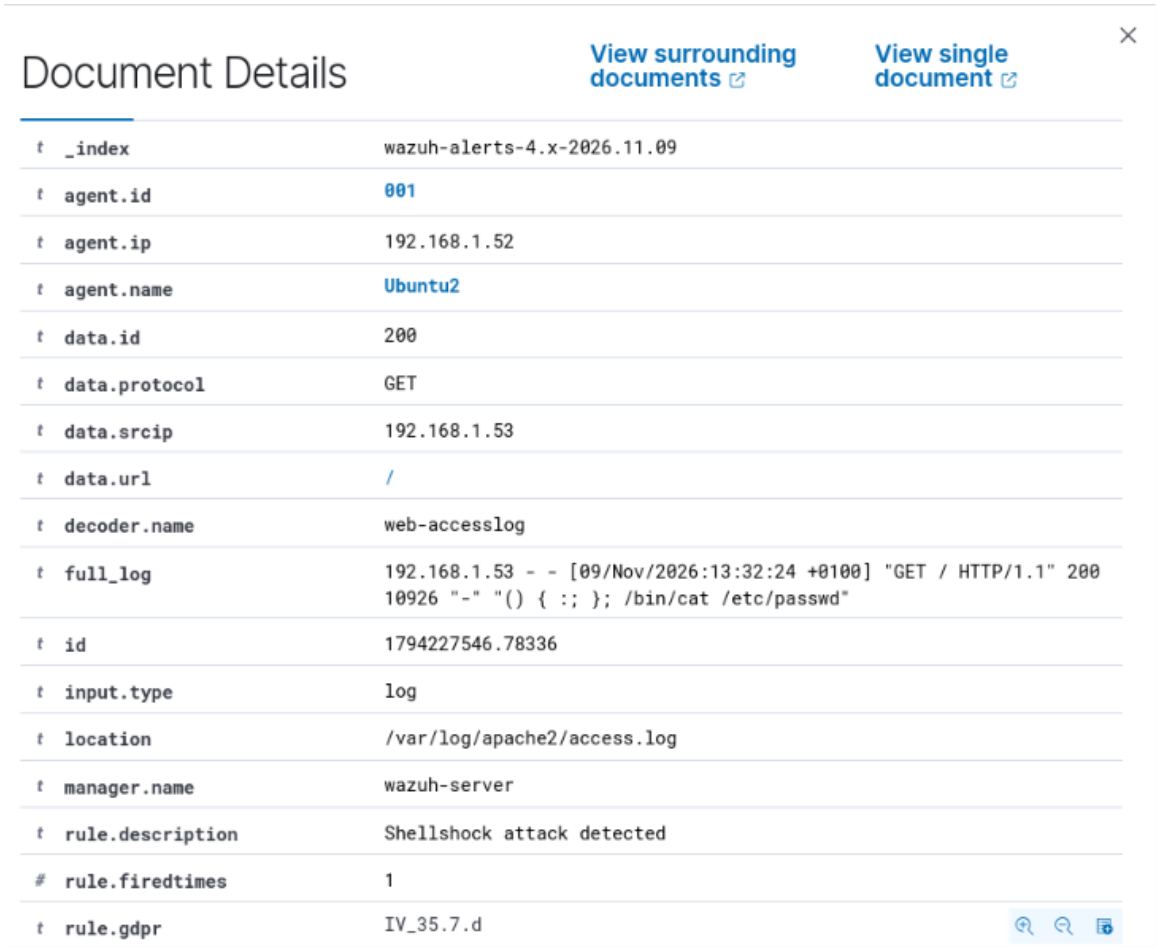



Figure 13. Wazuh Shellshock alert details, including source IP, payload, log location and rule.

## 6. Consolidated timeline of testing

| Date/time | Activity | Evidence / interpretation |
|---|---|---|
| **Before attack tests; exact time not shown** | Ubuntu endpoint health check and agent status review | Wazuh dashboard shows Ubuntu agent active; separate Ubuntu health-check screenshots supplied. Kali OS/network |

| | | |
|---|---|---|
| | | screenshots show Kali Rolling and 192.168.1.53. |
| **6 Nov 2026, 13:41:29.320–13:41:33.449** | Process-segfault events on Ubuntu2 | Wazuh event list shows process-segfault events shortly before Rootcheck detection. |
| **6 Nov 2026, 13:41:35.053** | Suspicious binary detection | Rootcheck detects trojaned /usr/bin/w; agent IP 192.168.1.55; rule 510, level 7. |
| **6 Nov 2026, 13:41:36.511 and 13:43:23.899** | PAM login-session events | Shown in the adjacent Wazuh event list; context and user activity should be validated. |
| **6 Nov 2026, 13:42:14–13:42:50.625** | File additions/deletions and integrity changes | Wazuh records file-added, file-deleted and checksum-changed events. One detailed record concerns /etc/cups/subscriptions.conf.N. |
| **6 Nov 2026, 20:44:49.823** | YARA/FIM malware-directory test | Modification alert for /tmp/yara/malware/mirai; before/after SHA-256 values differ. |
| **6 Nov 2026, 20:44:54–20:44:55** | Additional malware-directory file activity | FIM view shows added/modified entries including webshell, vpn_filter and xbash. |
| **9 Nov 2026, 13:32:24 (+0100)** | Shellshock-pattern HTTP request | Apache access log records source 192.168.1.53, GET /, payload attempting /bin/cat /etc/passwd, status 200. |
| **9 Nov 2026, 13:32:26.863** | Wazuh Shellshock alert | Rule 31168, level 15; target agent Ubuntu2 at 192.168.1.52. |
| **9 Nov 2026, 13:33:43.329** | Listened-port status change | Wazuh event list records a port-status change after the Shellshock alert. |

## 7. Cleanup and restoration evidence

The supplied restoration screenshots show Wazuh file-integrity events on 6 November 2026. At 13:42:50.625, agent Ubuntu2 (192.168.1.55) generated a 'File deleted' event for /etc/cups/subscriptions.conf.N. The detailed event records the process name /usr/sbin/cupsd, parent process /usr/lib/systemd/systemd, user root (UID 0), event 'deleted', and file mode 'whodata'. The screenshot also shows a checksum-change event at 13:42:50.625 and file modification time 13:42:45.000. A separate event list shows earlier file additions, deletions and checksum changes around 13:42:14–13:42:50.

Important limitation: these records demonstrate filesystem activity, but the deleted CUPS configuration file is not the trojaned /usr/bin/w binary or the YARA sample. The screenshots therefore cannot be used as proof that either malicious artefact was removed or restored. No post-restoration hash for /usr/bin/w, package verification result, clean YARA scan, or final endpoint-health screenshot is supplied.

Table JSON

| | |
|---|---|
| @timestamp | Nov 6, 2026 @ 13:42:50.625 |
| _index | wazuh-alerts-4.x-2026.11.06 |
| agent.id | 001 |
| agent.ip | 192.168.1.55 |
| agent.name | Ubuntu2 |
| decoder.name | syscheck_deleted |
| full_log | File '/etc/cups/subscriptions.conf.N' deleted<br>Mode: whodata |
| id | 1793968970.48563 |
| input.type | log |
| location | syscheck |
| manager.name | wazuh-server |
| rule.description | File deleted. |
| rule.firedtimes | 3 |
| rule.gdpr | II_5.1.f |
| rule.gpg13 | 4.11 |
| rule.groups | ossec, syscheck, syscheck_entry_deleted, syscheck_file |
| rule.hipaa | 164.312.c.1, 164.312.c.2 |

Figure 14. Wazuh restoration-related event: file deletion recorded for /etc/cups/subscriptions.conf.N.

W. Discover wazuh-alerts-4.x-2026.11.06#oATisKEBv2bBe9nNK-gi

| | |
|---|---|
| syscheck.audit.process.name | /usr/sbin/cupsd |
| syscheck.audit.process.parent_cwd | / |
| syscheck.audit.process.parent_name | /usr/lib/systemd/systemd |
| syscheck.audit.process.ppid | 1 |
| syscheck.audit.user.id | 0 |
| syscheck.audit.user.name | root |
| syscheck.event | deleted |
| syscheck.gid_after | 7 |
| syscheck.gname_after | lp |
| syscheck.inode_after | 1484677 |
| syscheck.md5_after | d75a7d2dc440a22b27976029f276f780 |
| syscheck.mode | whodata |
| syscheck.mtime_after | Nov 6, 2026 @ 13:42:45.000 |
| syscheck.path | /etc/cups/subscriptions.conf.N |
| syscheck.perm_after | rw-r----- |
| syscheck.sha1_after | fbd3638c818382276593ea1847bb01ab2667f688 |
| syscheck.sha256_after | 1399c52491ce0fdbe7360f26dc028bdbeba85809f79e5698db60aab04f127d98 |
| syscheck.size_after | 2,346 |
| syscheck.uid_after | 0 |

Figure 15. Detailed file-integrity event showing process, root user, deletion state and file metadata.

## 8. Required cleanup verification before closing

- Preserve forensic copies and hashes of the suspicious /usr/bin/w binary and the YARA sample.
- Quarantine or remove the test artefacts using the authorised lab procedure.
- Restore /usr/bin/w from a trusted package source if binary tampering is confirmed.
- Run package-integrity verification and record the result.
- Run YARA again and retain the rule name, scan output and final file hashes.
- Review authentication, process, Apache and system logs for follow-on activity.
- Repeat the endpoint health check and capture the Wazuh agent status after remediation.
- Record the operator, exact cleanup time, commands/actions taken and verification outcome.

## 9. Overall conclusion

The project demonstrates Wazuh monitoring across file integrity, Rootcheck and web access-log detection. The YARA-related test produced file-modification evidence, the suspicious-binary test generated a Rootcheck alert for /usr/bin/w, and the Shellshock test generated a high-severity alert for a request from the Kali host to the Ubuntu web server. The strongest confirmed result is the Shellshock pattern match because the alert includes the request payload, source and target addresses, log location and rule ID. The endpoint should not be declared fully restored until the affected binary and malware test artefacts have been verified clean and a post-cleanup health check has been captured.